

NENSI BAI

Graduate Trainee Engineer | Software Operations & Data Automation

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SUMMARY

Detail-oriented Graduate Trainee Engineer with 1 year of professional experience specialising in software operations, data automation, and backend development. Technical expertise includes Python, Java, Spring Boot, and database management. Proven track record of developing enterprise systems, optimising backends, and executing technical data validation.

TECHNICAL SKILLS

Programming & Frameworks	Python, Java, Spring Boot, JPA, SQL, HTML, CSS
Data & Automation	MySQL, Pandas, NumPy, Data Validation, Excel
Core Fundamentals	OOP, SDLC, Debugging, API Integration, Requirements Analysis
Tools & Platforms	Git, GitHub, Tkinter, Odoo (Basic)

PROFESSIONAL EXPERIENCE

Graduate Trainee Engineer — Software Operations & Data Automation April 2026 – Present
Utopia Industries

- Streamline software operations and implement data automation workflows to enhance enterprise system efficiency.
- Collaborate on backend logic monitoring, database tuning, and performance improvement tasks.

Trainee Engineer March 2026 – April 2026
PTP Pvt. Limited

- Managed core software operations tasks and performed systematic validation of cross-functional system behaviours.

Python Developer — Intern June 2025 – July 2025
Odolution

- Executed Python-based module customisation, backend debugging, and rigorous functional system validation workflows.

Machine Learning Intern September 2024 – October 2024
NCAI NCL

- Developed and optimised a concrete strength prediction model using ML algorithms to process and analyse raw data.

TECHNICAL PROJECTS

Enterprise Inventory Pro System *Java, Spring Boot, JPA, MySQL, GitHub*
Developed a robust backend architecture using Spring Boot and JPA for automated inventory tracking and secure transactional database controls.

Cleaning AI Robot (CLAIR) *Machine Learning, OpenCV, ESP32-CAM, Embedded Systems*
Evaluated real-environment embedded system behaviour, advanced object detection accuracy, and multivariable sensor integration arrays.

Sudoku Game Using Algorithms *Python, AI Algorithms, Backtracking, Tkinter*
Built an interactive algorithmic game app applying recursive backtracking logic and functional UI verification.

EDUCATION

NED University of Engineering & Technology, Karachi 2021 – 2025 | CGPA: 2.87
Bachelor of Engineering — Computer and Information Systems